[Total No. of Questions: 09]

Uni. Roll No. ....

[Total No. of Pages: 01]

Program/ Course: **B.Tech. (Sem. 4th)**

Name of Subject: **Computer Architecture and Microprocessors**

Subject Code: **IT-14405**

Paper ID: **15336**

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

1) **Section-A** is **compulsory**
2) Attempt any **four** questions from **Section-B** and any **two** questions from **Section-C**
3) Any missing data may be assumed appropriately

Section – A

Q1. **[Marks: 02 each]**

a) What is the word length of 8085?

b) Distinguish between computer architecture and computer organization.

c) What is the role of registers in computers?

d) Distinguish between direct and indirect address instructions.

e) What is meant by non maskable interrupt? Give an example of non maskable interrupt.

f) Compare the features of RISC and CISC.

g) Illustrate the purpose of address latch enable.

h) Differentiate between LDA and LDAX instructions.

i) What is the difference between MOV AX, 1000H and MOV AX, [1000H]?

j) Outline the basic difference between RIM and SIM.

Section – B

[Marks: 05 each]

Q2. Write a program to fund sum of series of 8 bit numbers.

Q3. Write a program to find subtraction of two 16 bit numbers.

Q4. Discuss the various types of addressing modes for 8085 microprocessor.

Q5. Illustrate the memory hierarchy in order of their features with their comparative analysis.

Q6. Elaborate the various types of data transfer operations done in assembly language programming

Section – C **[Marks: 10 each (05 for each sub-part, if any)]**

Q7. Describe with the help of a neat diagram, the working principle of DMA.

Q8. Explain with the help of block diagram architecture of 8085.

Q9. Write short notes on a) 32 bit microprocessor b) Applications of microprocessor.

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